PROJECT DOCUMENTATION:

LAUNCH YOUR LINUX MACHINE ON AWS WITH PUTTY





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🚀 PROJECT OVERVIEW

This project demonstrates how to deploy a secure Linux server on Amazon Web Services (AWS) EC2 and establish a remote connection using PuTTY on a Windows system. Throughout this hands-on implementation, you will learn how to create an EC2 instance, configure security settings, generate and manage SSH key pairs, convert the AWS private key for PuTTY compatibility, and securely access your cloud server. By completing this project, you will gain practical experience with cloud computing, Linux administration, and secure remote access—essential skills for aspiring Cloud and DevOps Engineers. ☁️🐧🔐

🧳 PREREQUISITES

Before you begin, ensure you have the following:

- **AWS Account** – An active AWS account with permission to create EC2 resources.
- 💻 **Windows Computer** – Required to connect using the PuTTY SSH client.
- 🔑 **AWS Key Pair (.pem)** – Generated during EC2 instance creation for secure authentication.🛠️
- **PuTTY** – SSH client used to connect to the Linux instance.
- 🔄 **PuTTYgen** – Utility to convert the .pem key into a .ppk format compatible with PuTTY.
- 🌐 **Stable Internet Connection** – Required to communicate with the AWS cloud instance.
- 🐧 **Basic Linux Knowledge** – Familiarity with Linux commands and terminal navigation is helpful.
- 🔒 **Understanding of SSH** – Basic knowledge of Secure Shell (SSH) for remote server access.

⌨ **Command Line Familiarity**: Comfortable using basic Linux commands (e.g., ls, cd, pwd).

Once you've got these, you're all set to launch your AWS Linux instance and connect using PuTTY 🚀

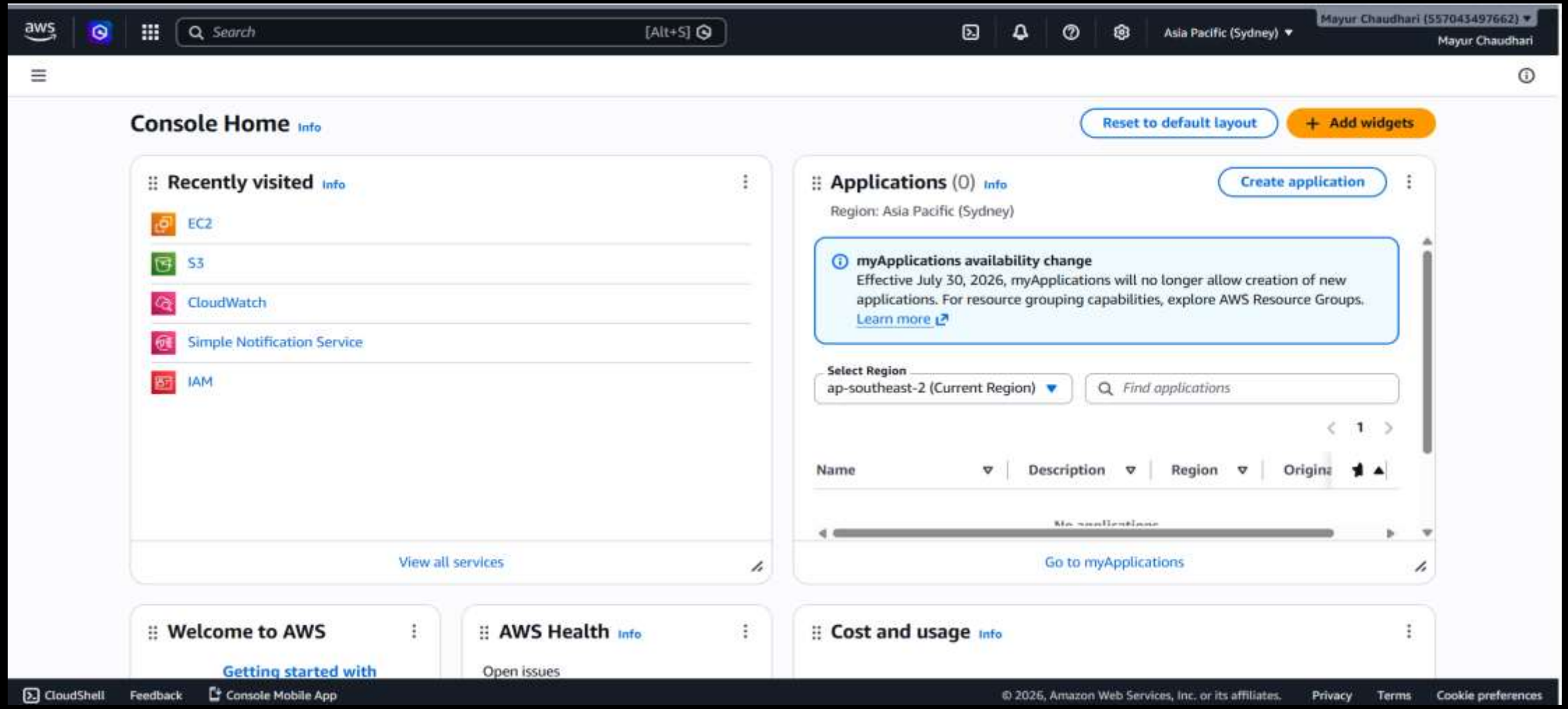
WHAT IS PUTTY? 💻

PuTTY is a free and open-source SSH client that allows you to securely connect to remote servers, typically Linux-based, from a Windows machine. It enables you to access and manage your server via the command line.

Ready to get your Linux server running on AWS and connect to it using PuTTY ?
Let's go through the steps together in a fun and easy way 🚀

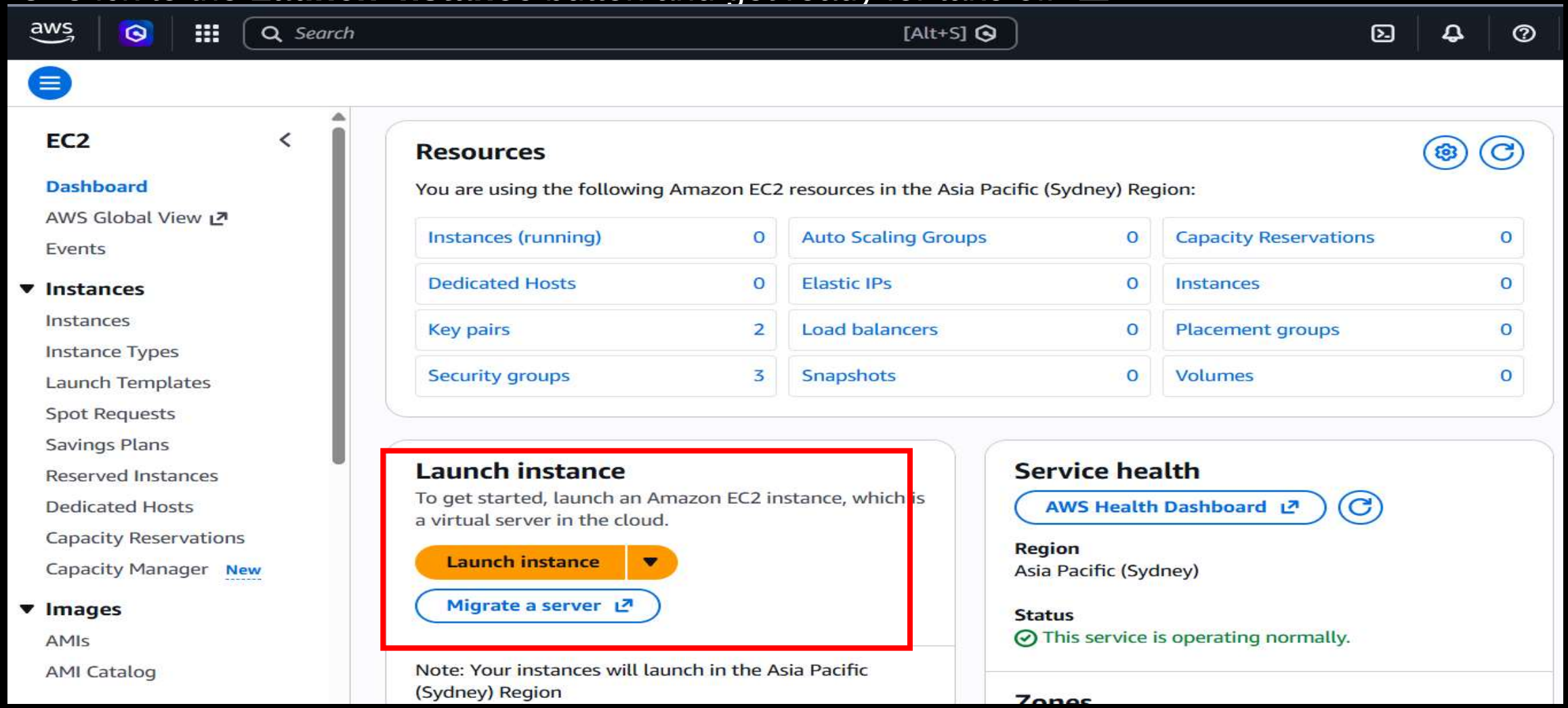
I. LAUNCH YOUR EC2 INSTANCE ON AWS 🚀

1.Log in to your AWS Management Console. It's time to get started! 🚀



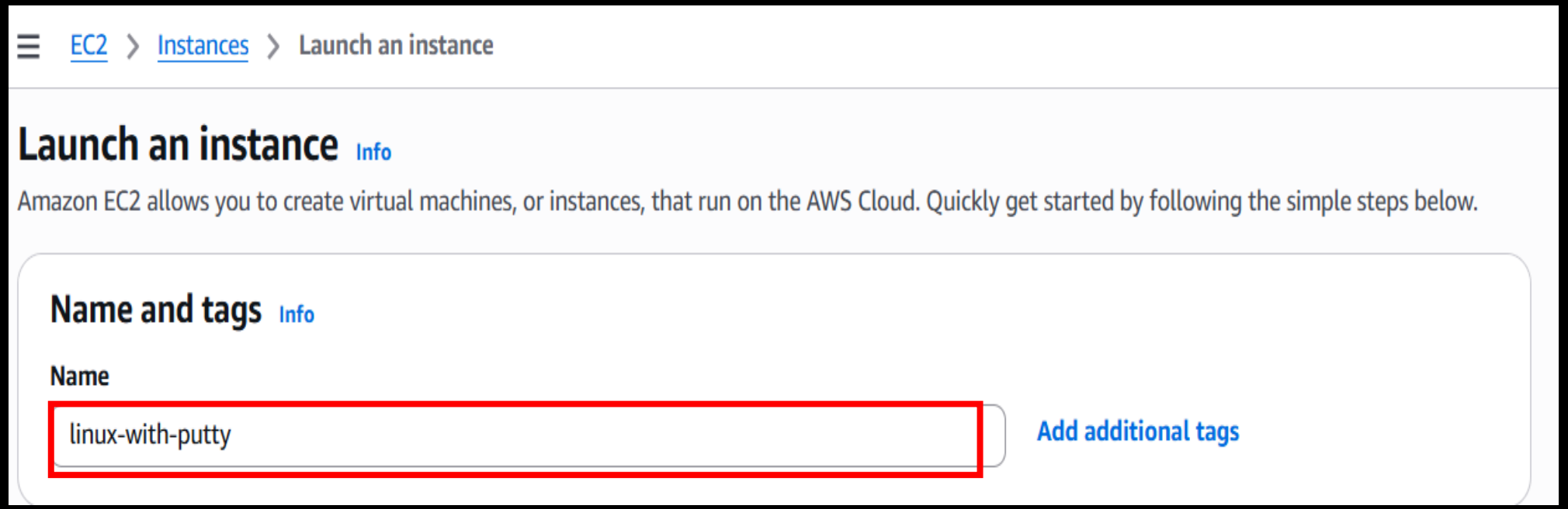
2. Head Over to **EC2 Dashboard** Under **Services**, the heart of your cloud adventure.

3. Click to the **Launch Instance** button and get ready for take off. ✈

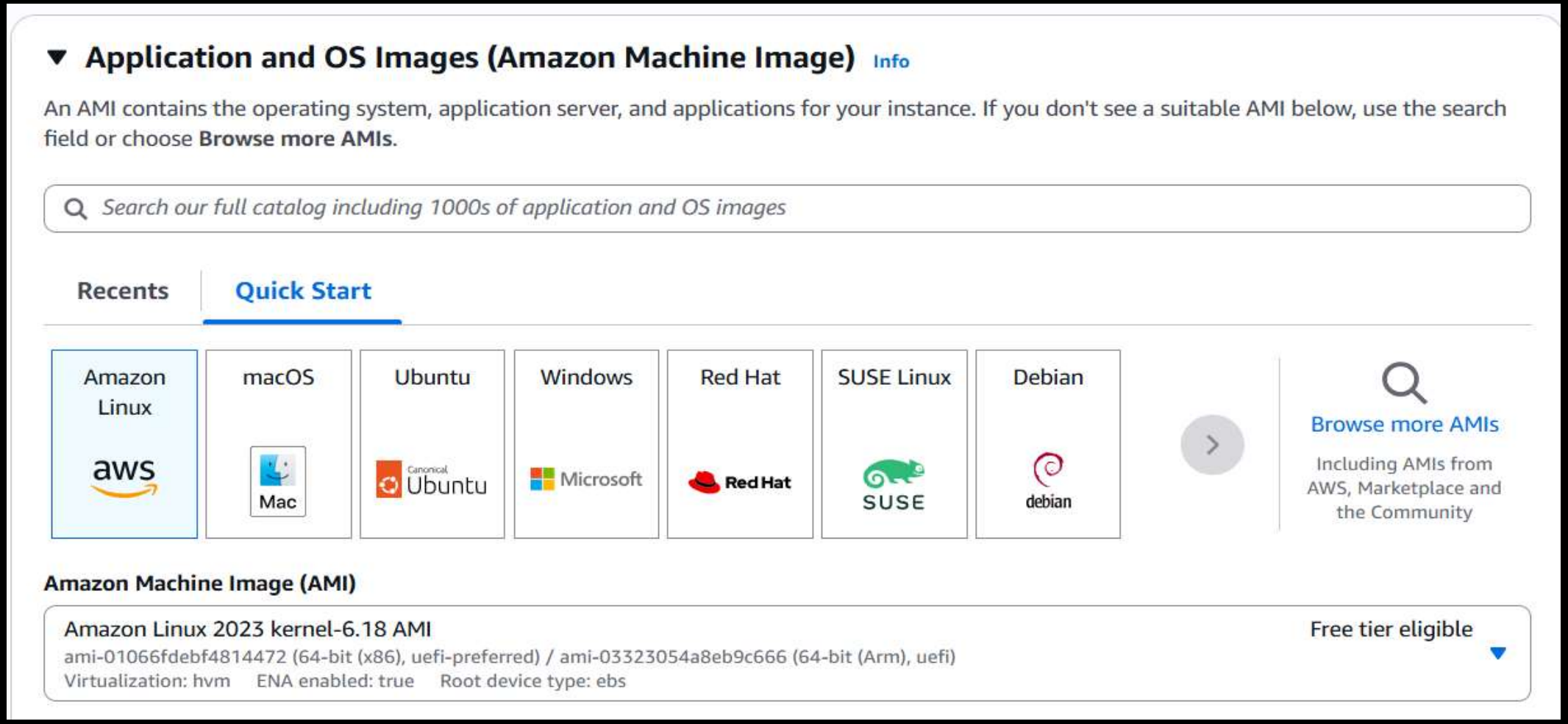


4. **Give a name** to your instance, this helps you easily identify it later. ✓

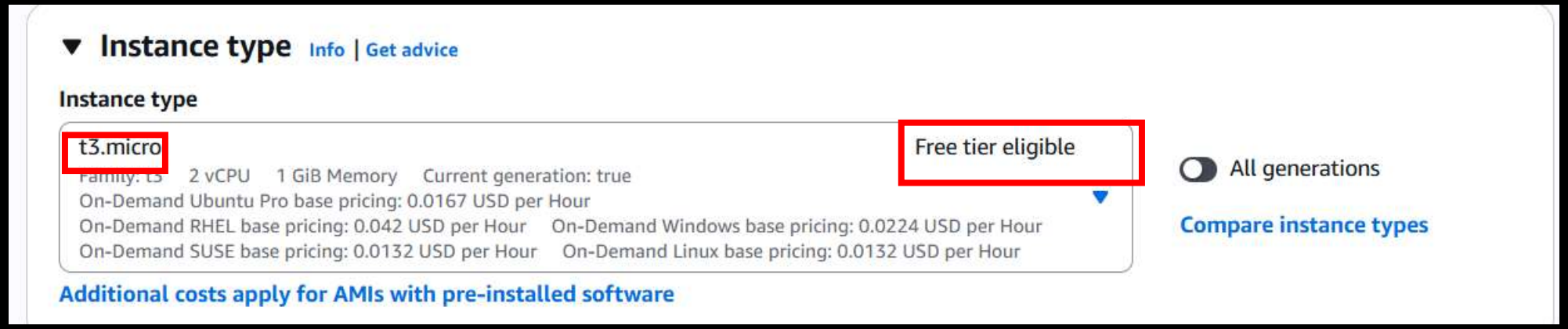
(**Eg- linux-with-putty**)



5. Select your **Amazon Machine Image(AMI)** (eg., **Red Hat or Amazon Linux**).



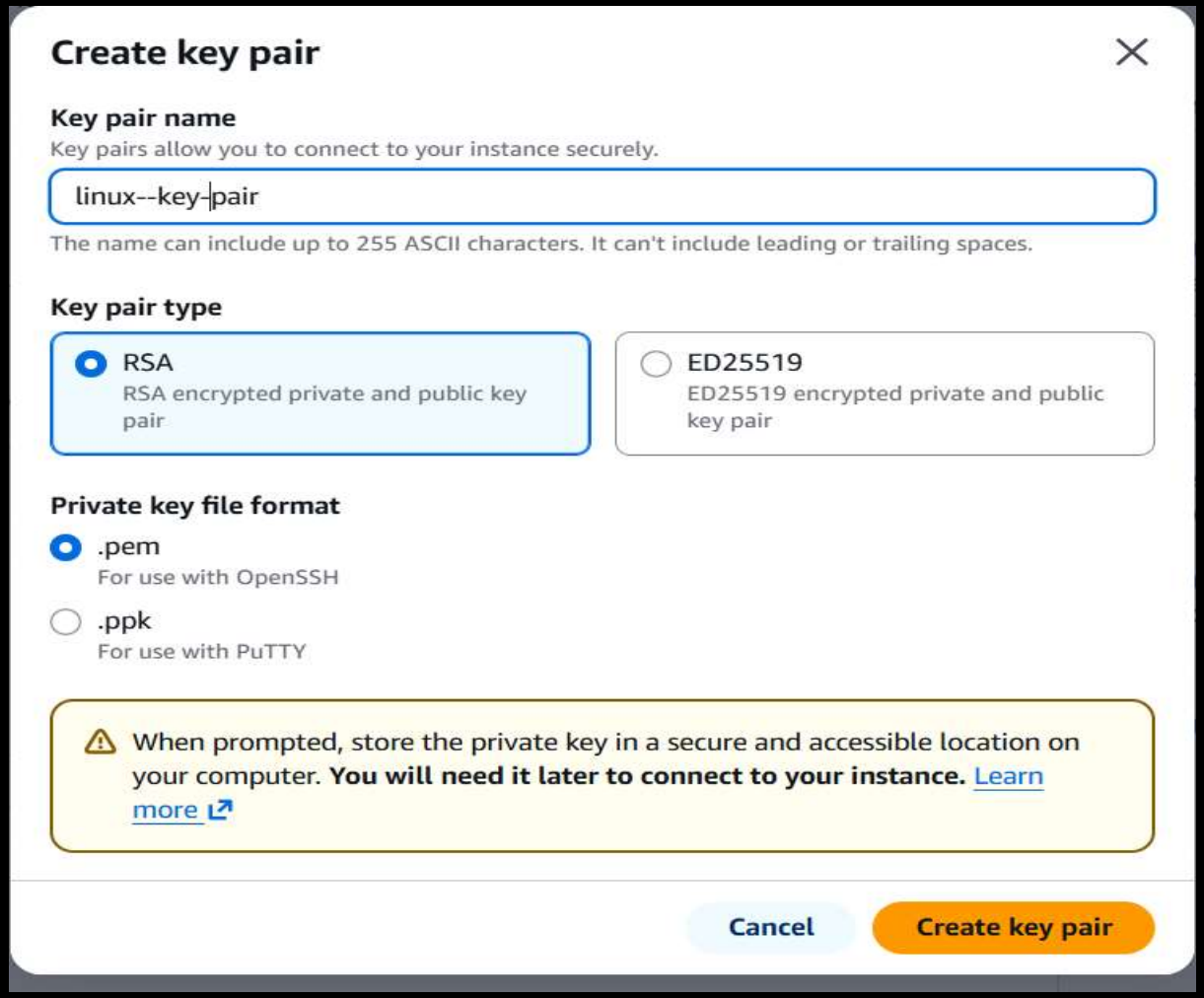
6. Choose an Instance Type: select **t3.micro,**which is perfect for beginners and falls within the free tier offering 🎁



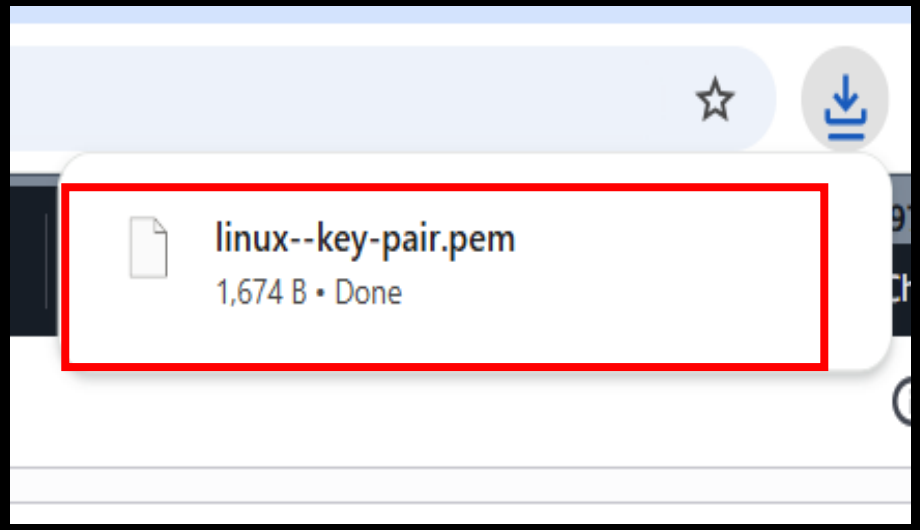
7. In the **Key pair** section, click on **Create a new pair** to generate a new key pair for secure access.🔐



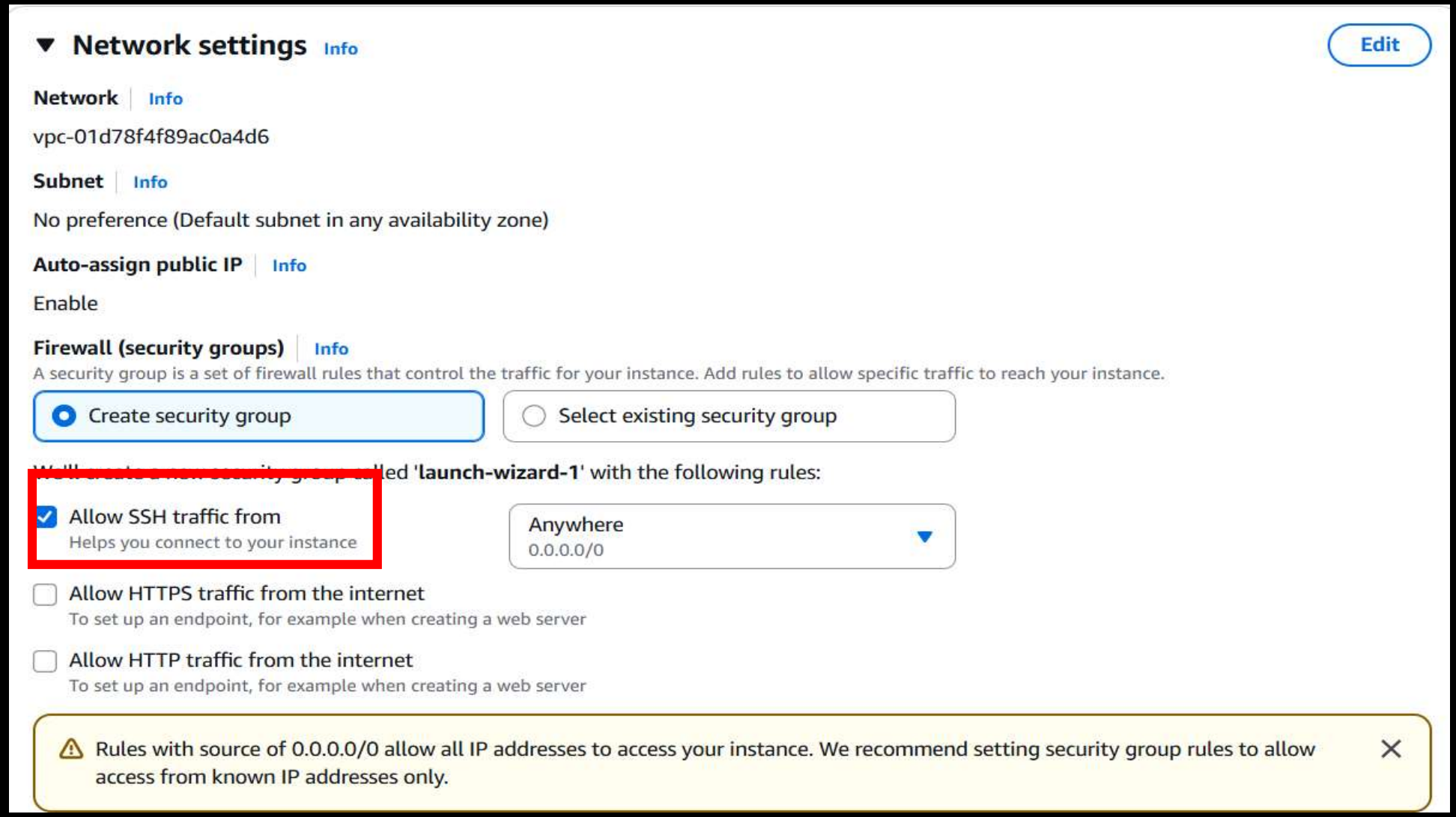
8. Give Your Key Pair a name (eg., linux-key-pair). 📜



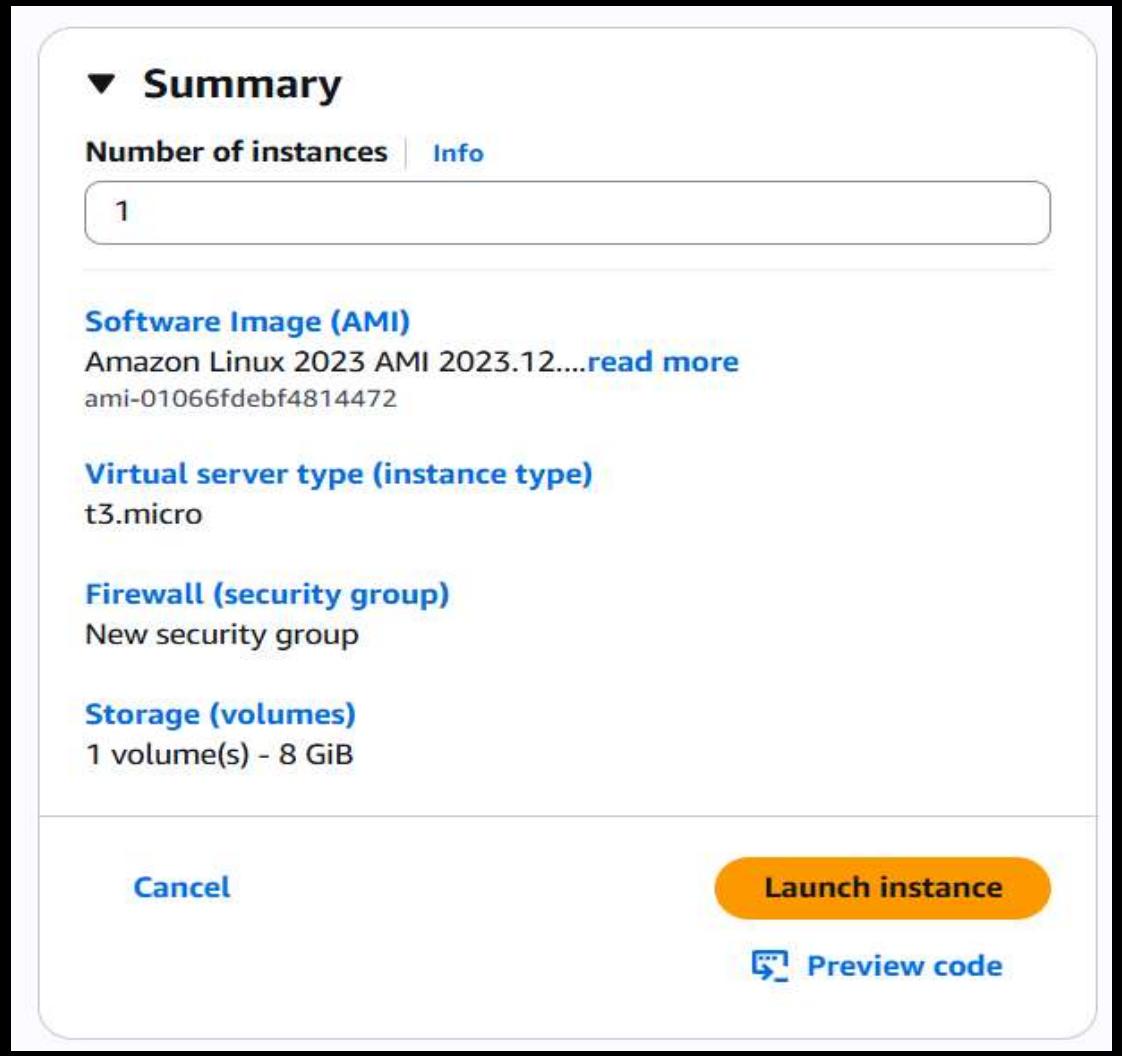
9. Hit **Create Key Pair** and don't forget to **Download the private key(.pem file)** when prompted. You'll need this to access your server securely! 🔑



10.Configure your **Security Group**, Just make sure **SSH** (port 22) is allowed.🔐



11.See the instance **Summary** and click on **Launch Instance**



➢ What is PuTTYgen ?🔑

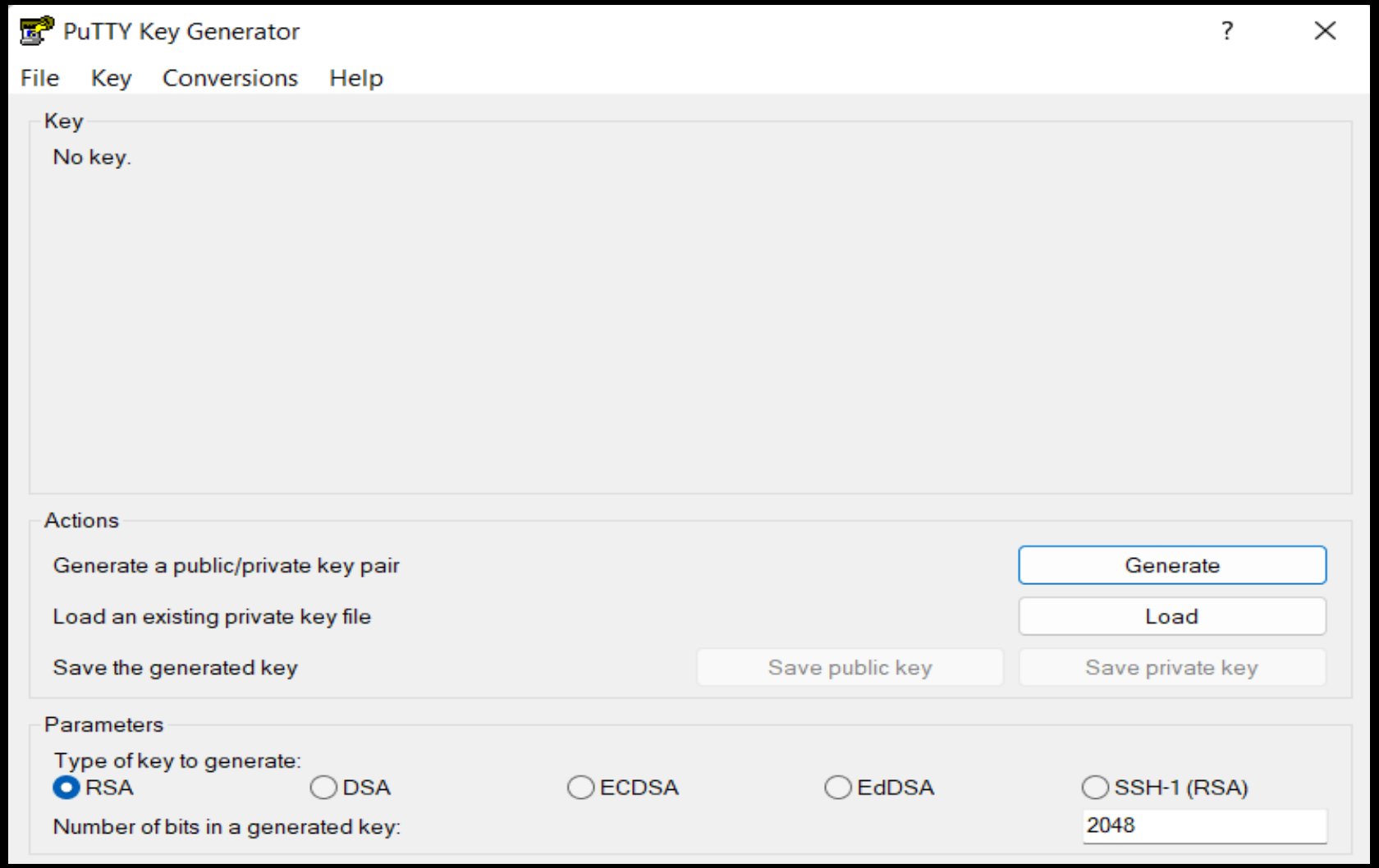
PuTTYgen is a tool used to generate and convert SSH key pairs. It allows you to convert AWS EC2 key pairs(.pem files) into PuTTY-compatible format (.ppk),
Which is required for securely connecting to your server via PuTTY..🔐💻

➢ Why We Use PuTTYgen?

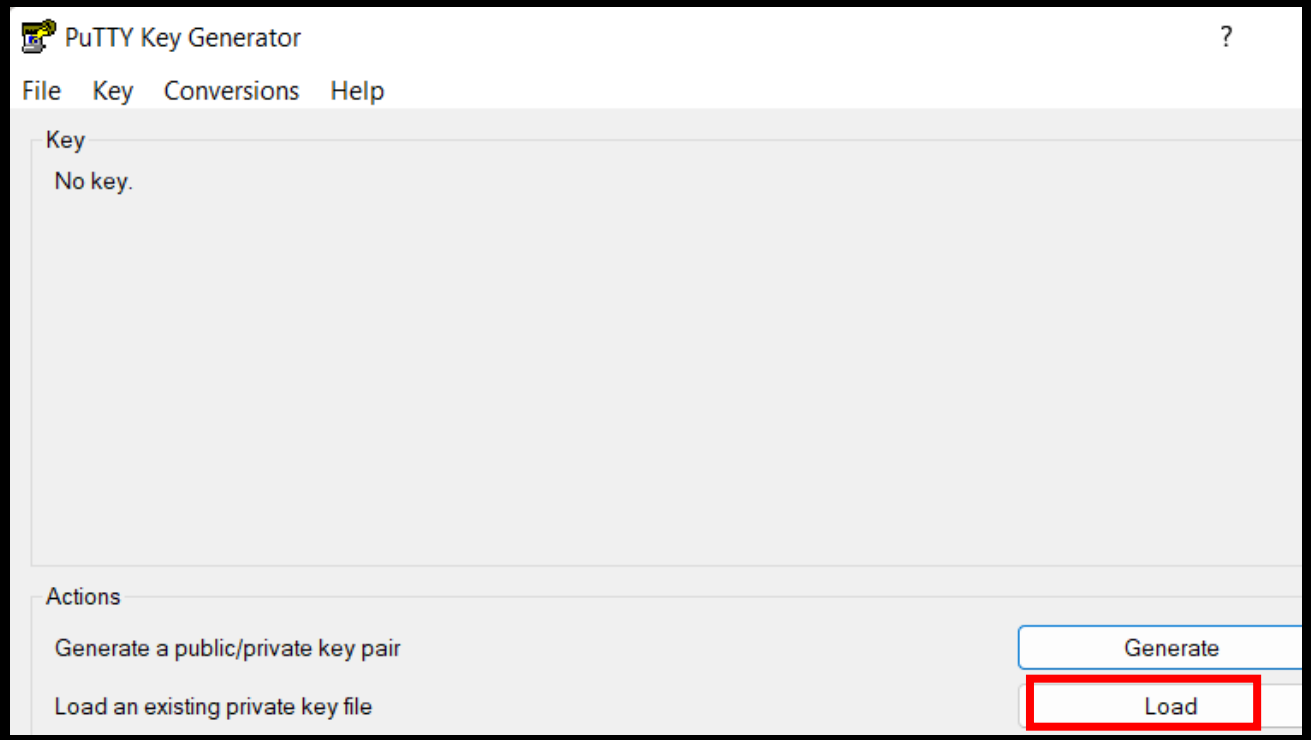
We use PuTTYgen to convert AWS EC2 key pairs (.pem files) into a PuTTY-compatible Format(.ppk),allowing us to securely connect to a remote Linux server using PuTTY. Without this conversion, PuTTY cannot authenticate the connection.🔐💻

➢ **CONVERT YOUR .PEM FILE TO .PPK USING PUTTYGEN** 🔑

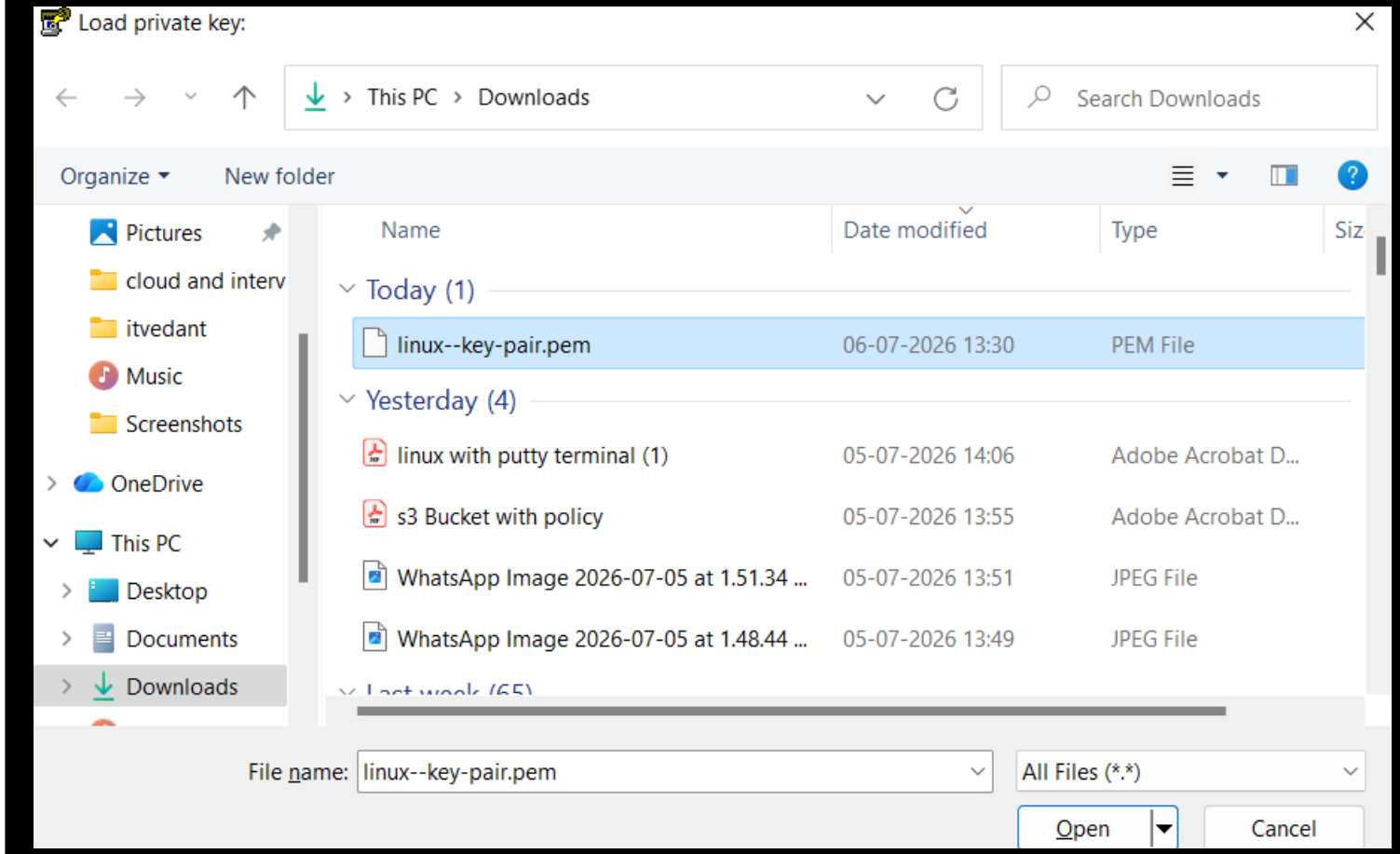
1.Open PuTTYgen , your magical key converter. ✨

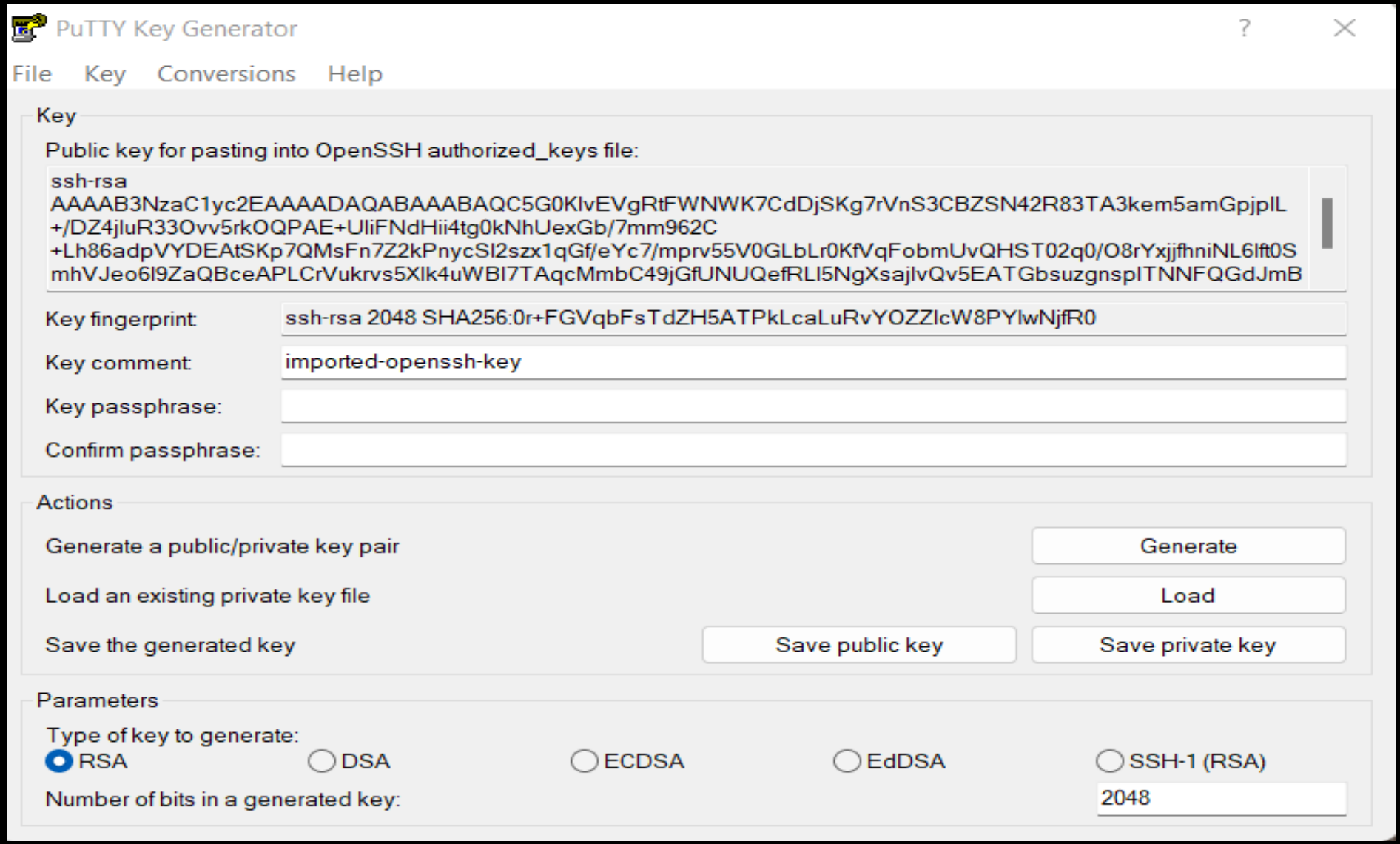

PuTTY Key Generator
File
Key
Conversions
Help
Key
No key.
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048

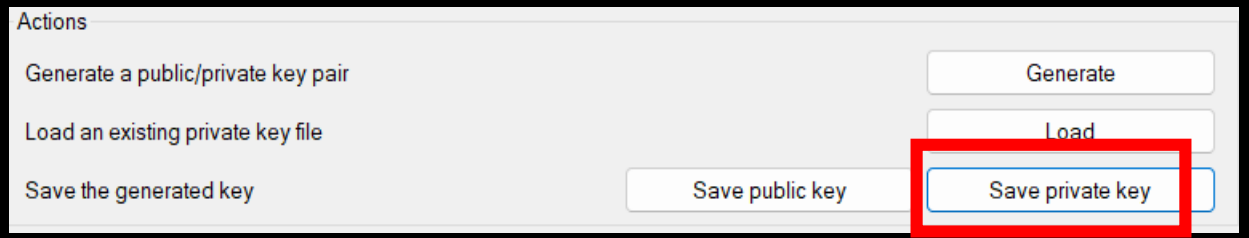
2. Click **Load** and select your freshly downloaded **.pem file.** 🔓

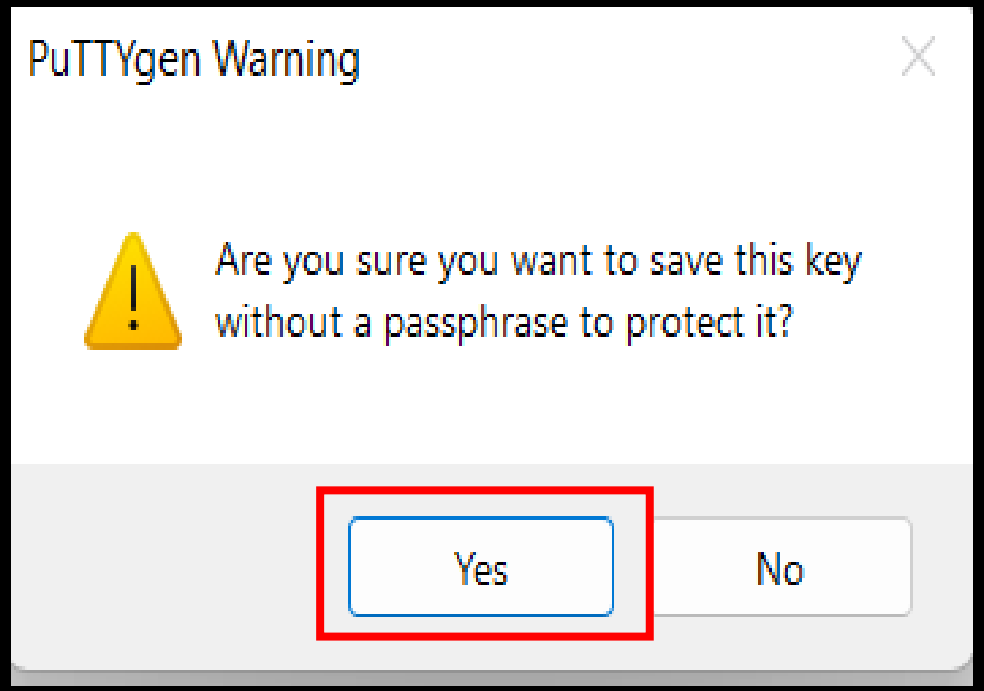


3.Select the **key .pem file.**

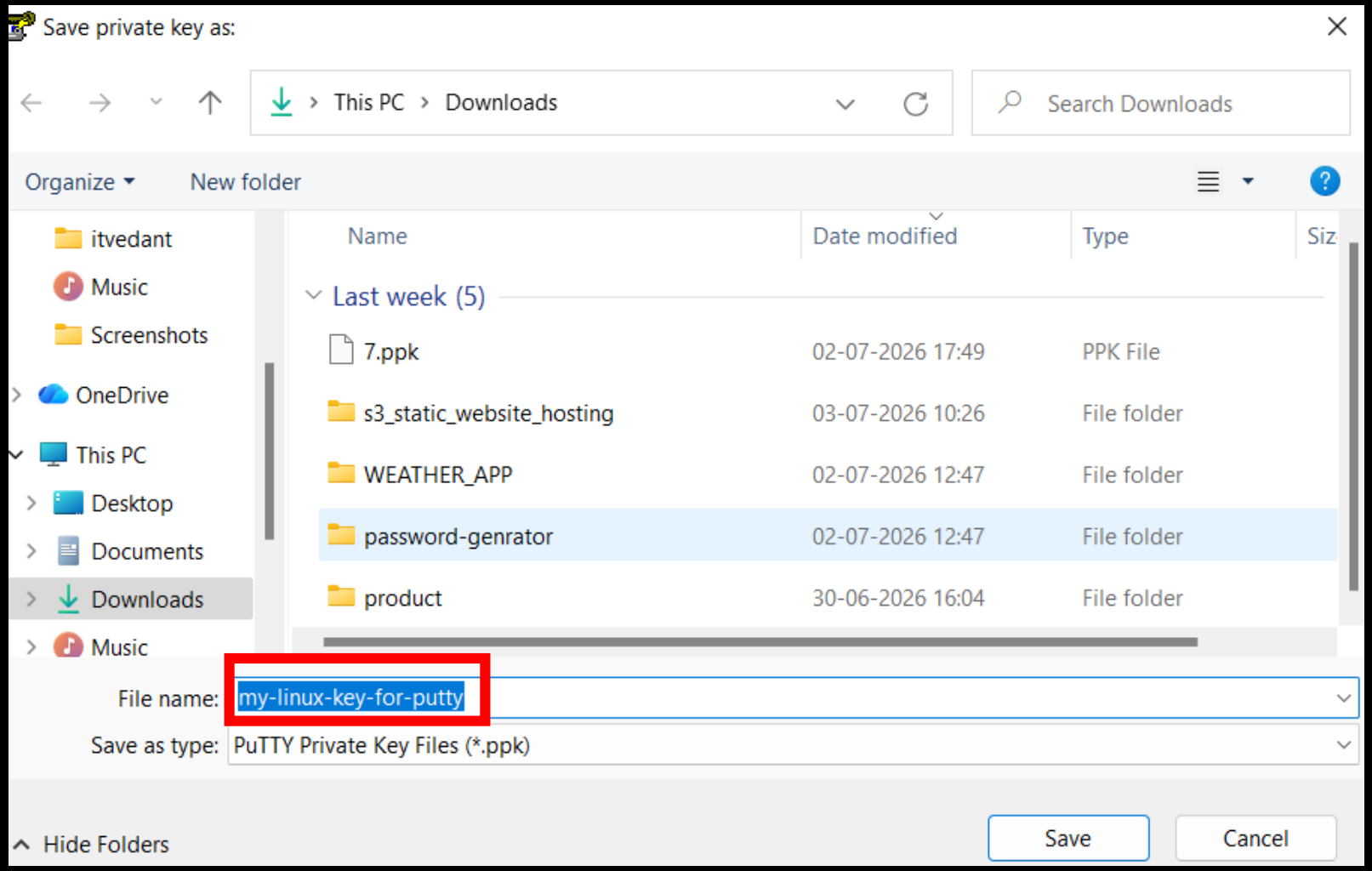



PuTTY Key Generator
File
Key
Conversions
Help
Key
Public key for pasting into OpenSSH authorized_keys file:
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQC5G0KlvEVgRtFWNWK7CdDjSKg7rVnS3CBZSN42R83TA3kem5amGpjplL
+/DZ4jluR33Ovv5rkOQPAE+UliFNdHii4tg0kNhUexGb/7mm962C
+Lh86adpVYDEAtSKp7QMsFn7Z2kPnycSl2szx1qGf/eYc7/mprv55V0GLbLr0KfVqFobmUvQHST02q0/O8rYxjjfhniNL6lft0S
mhVJeo6l9ZaQBceAPLCrVukrvs5Xlk4uWBI7TAqcMmbC49jGfUNUQefRLI5NgXsajlvQv5EATGbsuzgnsplTNNFQGdJmB
Key fingerprint:
ssh-rsa 2048 SHA256:0r+FGVqbFsTdZH5ATPkLcaLuRvYOZZlcW8PYlwNjfR0
Key comment:
imported-openssh-key
Key passphrase:
Confirm passphrase:
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048


Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key


PuTTYgen Warning
Are you sure you want to save this key without a passphrase to protect it?
Yes
No

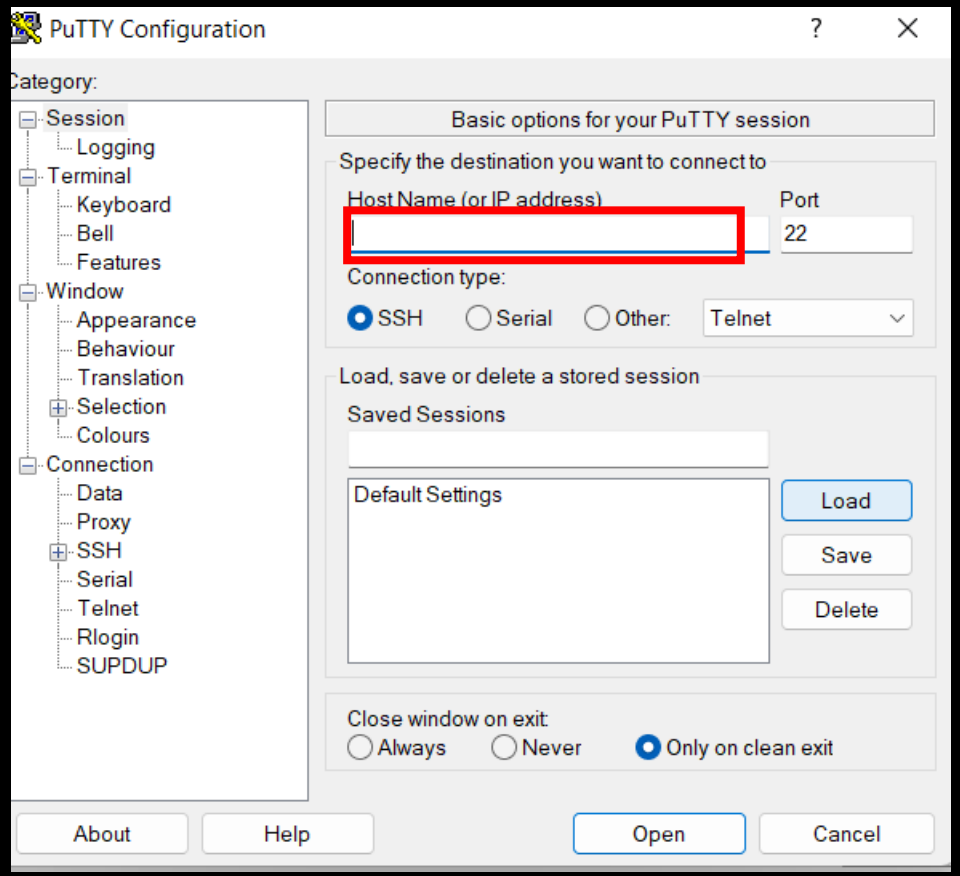
3.Save the private key as a **.ppk file** (this is PuTTY's preferred format). 🔑

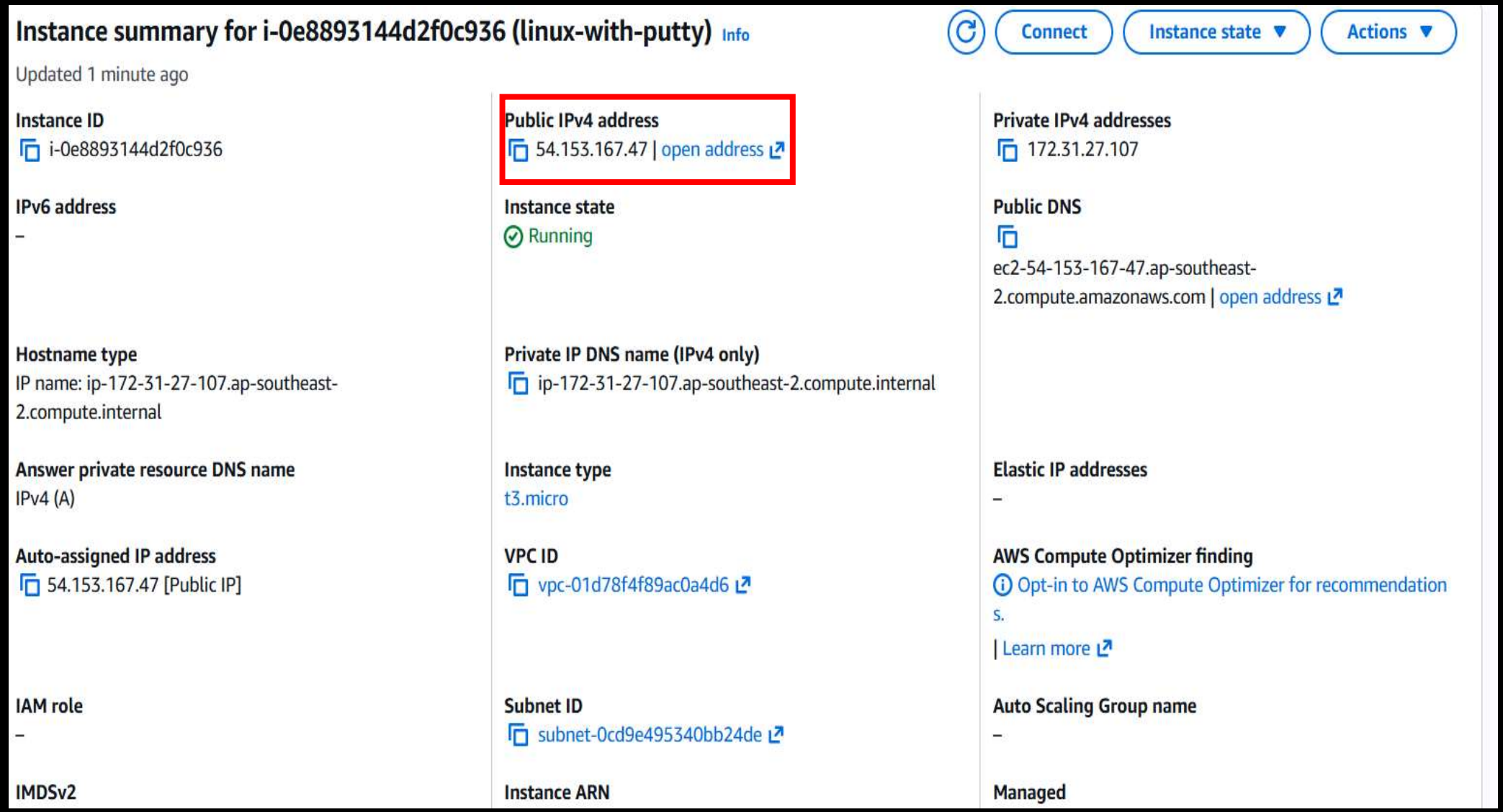


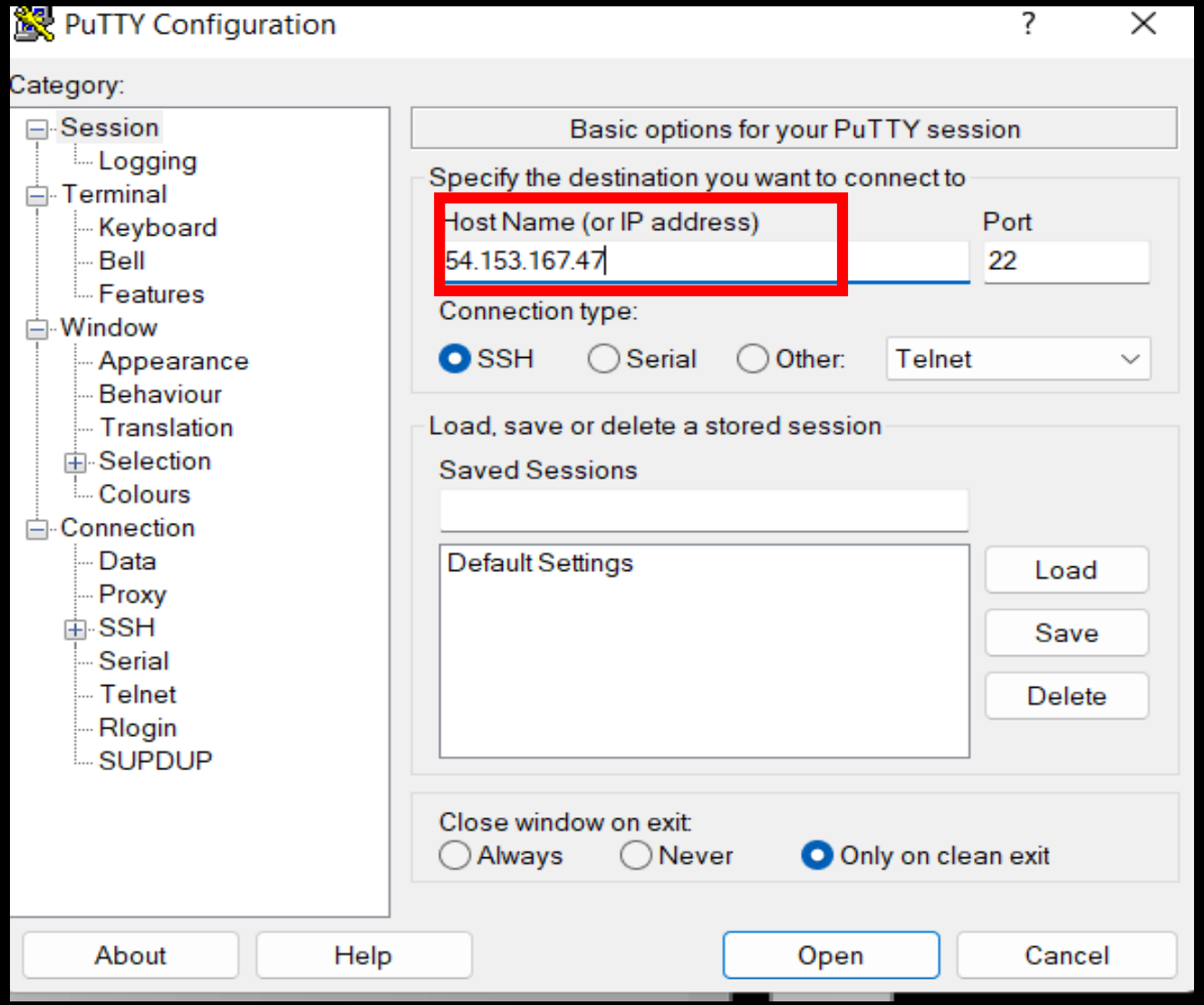
➢ CONNECT TO YOUR EC2 INSTANCE WITH PUTTY 💻

1.Open PuTTY and get ready to teleport into your EC2 instance. 💻

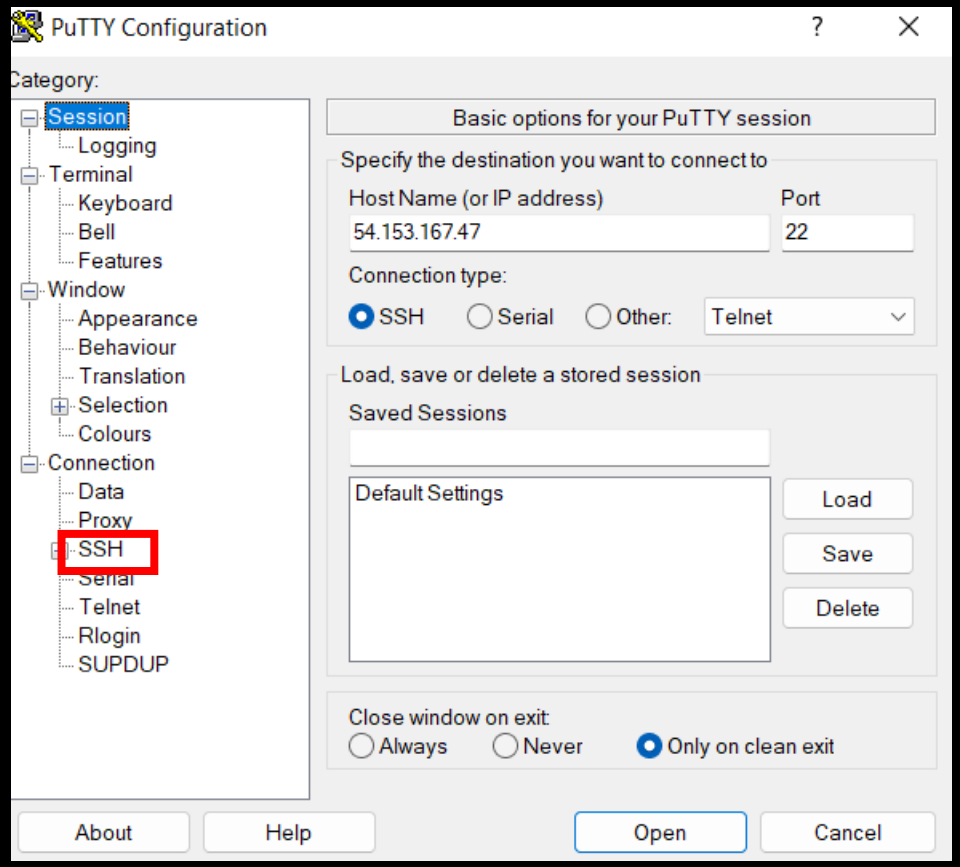
2.In the Host Name field, enter the Public IP Address or Public DNS of your EC2 instance.🚀

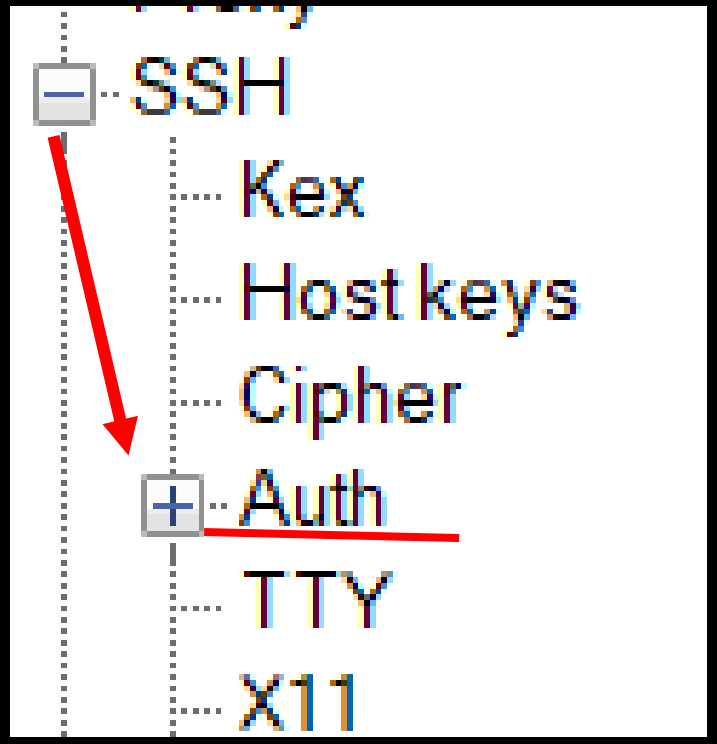


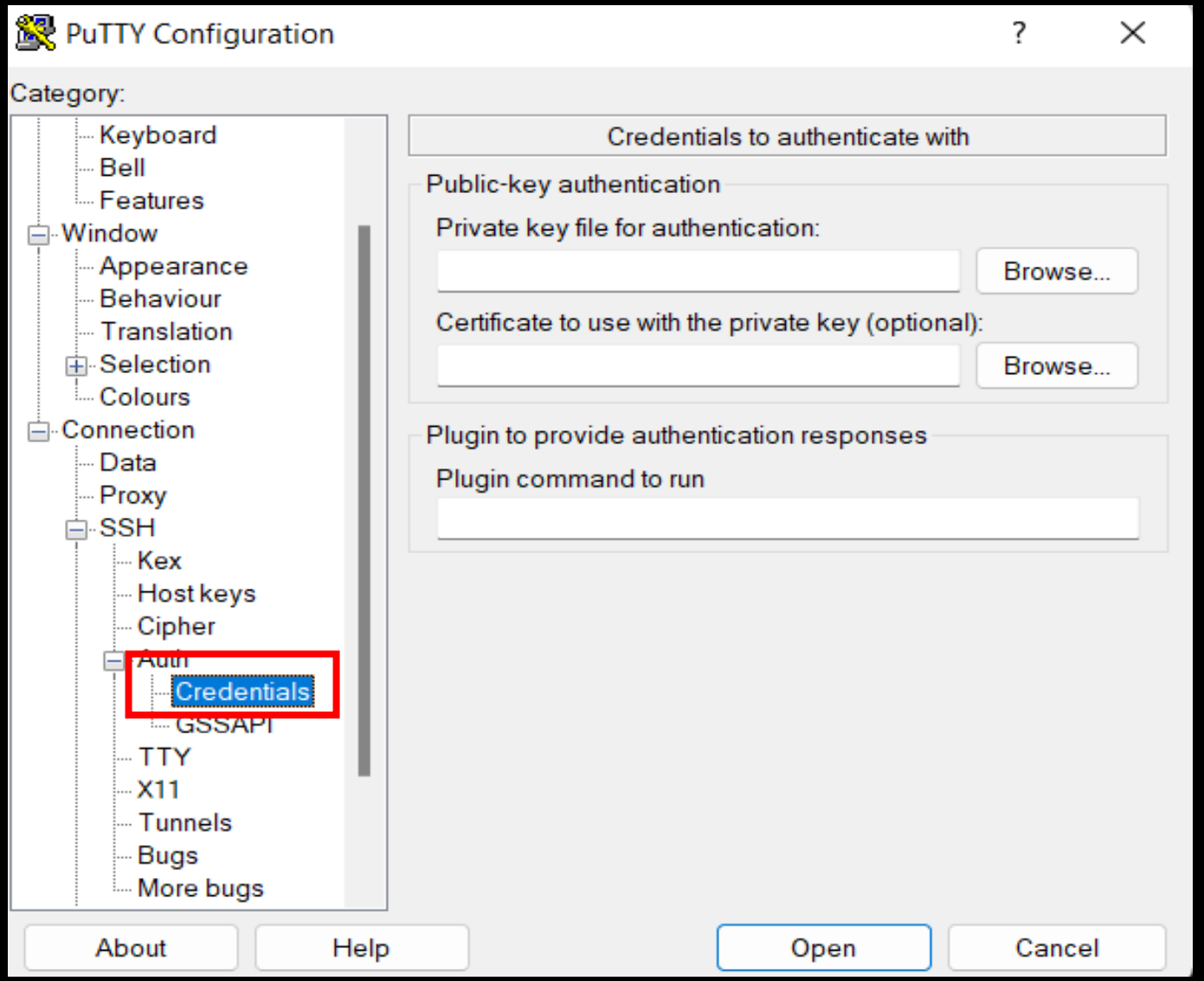

Instance summary for i-0e8893144d2f0c936 (linux-with-putty) Info
Updated 1 minute ago
Connect
Instance state
Actions
Instance ID
i-0e8893144d2f0c936
Public IPv4 address
54.153.167.47 | open address
Private IPv4 addresses
172.31.27.107
IPv6 address
–
Instance state
Running
Public DNS
ec2-54-153-167-47.ap-southeast-2.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-27-107.ap-southeast-2.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-27-107.ap-southeast-2.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
54.153.167.47 [Public IP]
VPC ID
vpc-01d78f4f89ac0a4d6
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more
IAM role
–
Subnet ID
subnet-0cd9e495340bb24de
Auto Scaling Group name
–
IMDSv2
Instance ARN
Managed


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
54.153.167.47
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel

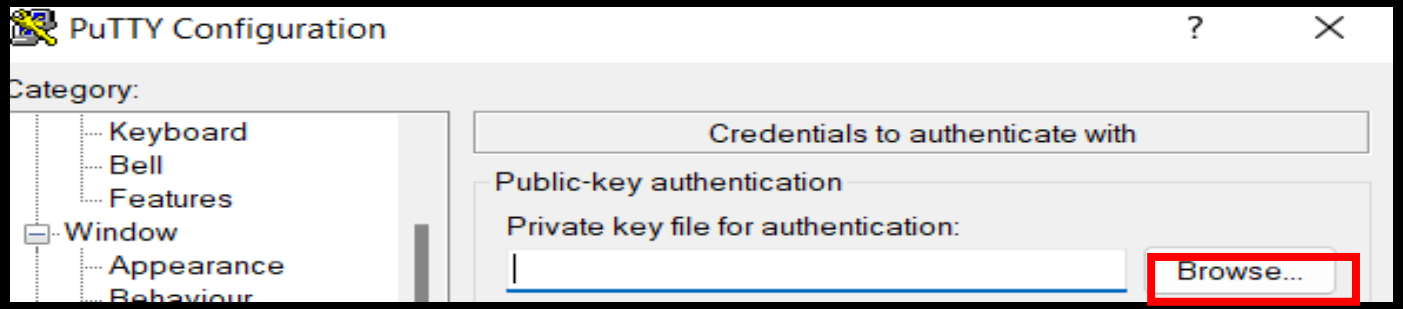
3.Go to **SSH >Auth** on the left Menu. 🗁

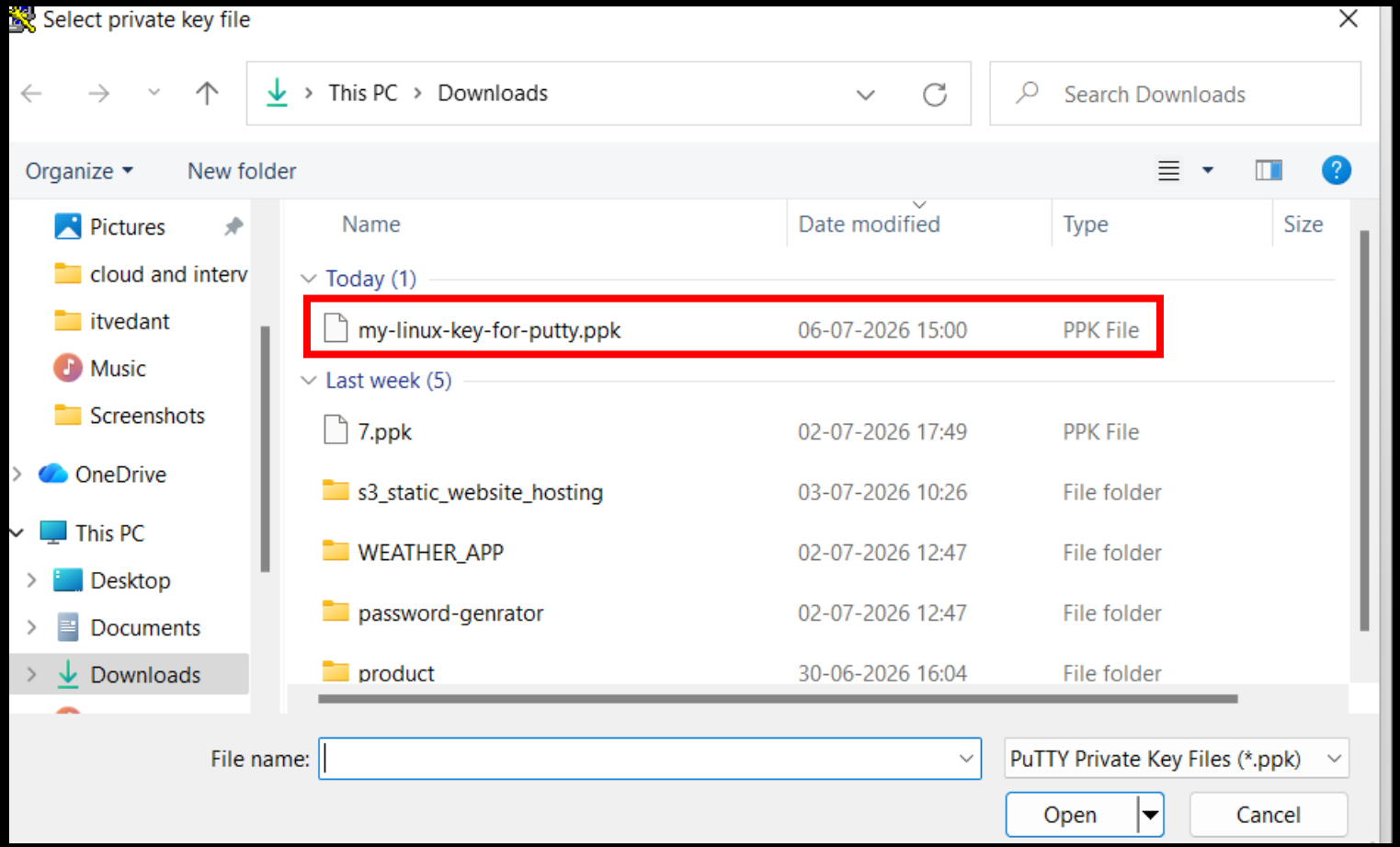




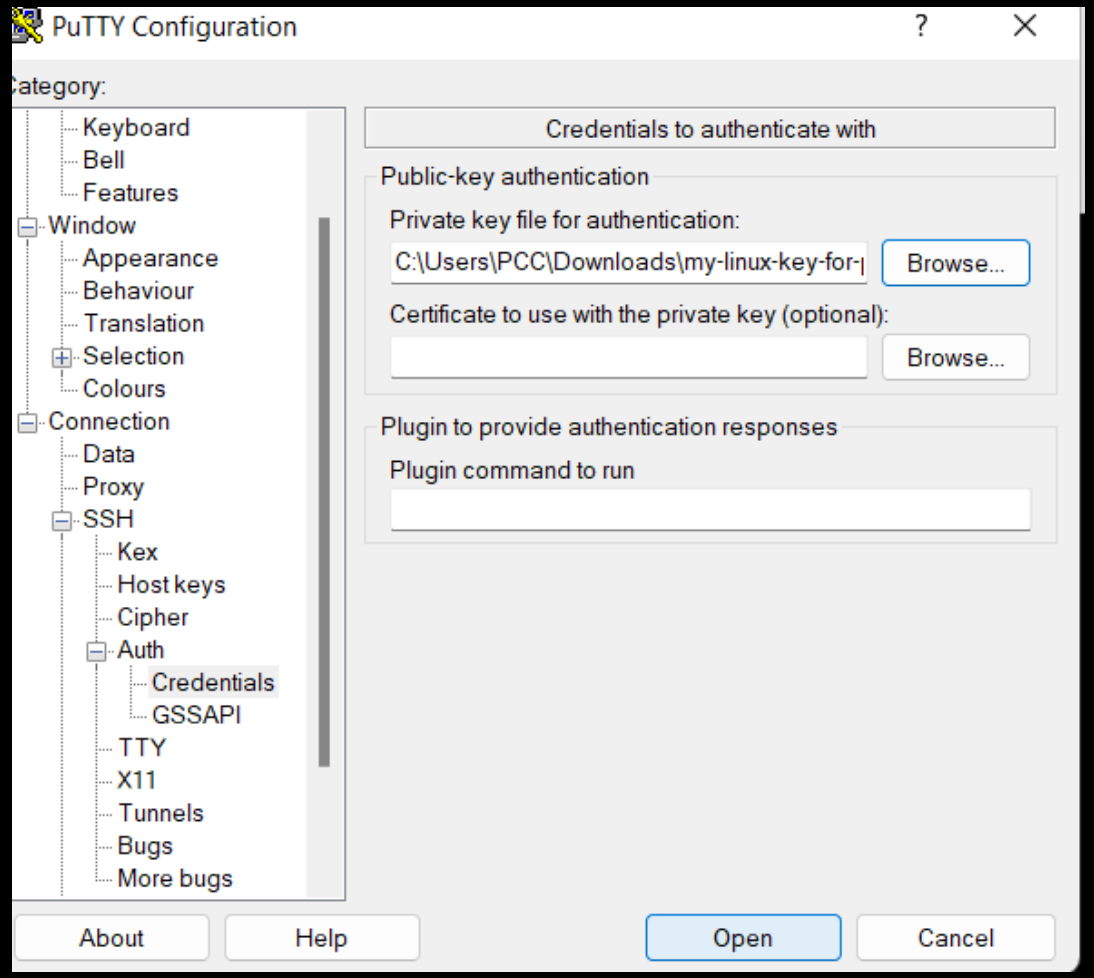

PuTTY Configuration
Category:
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Bugs
More bugs
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel

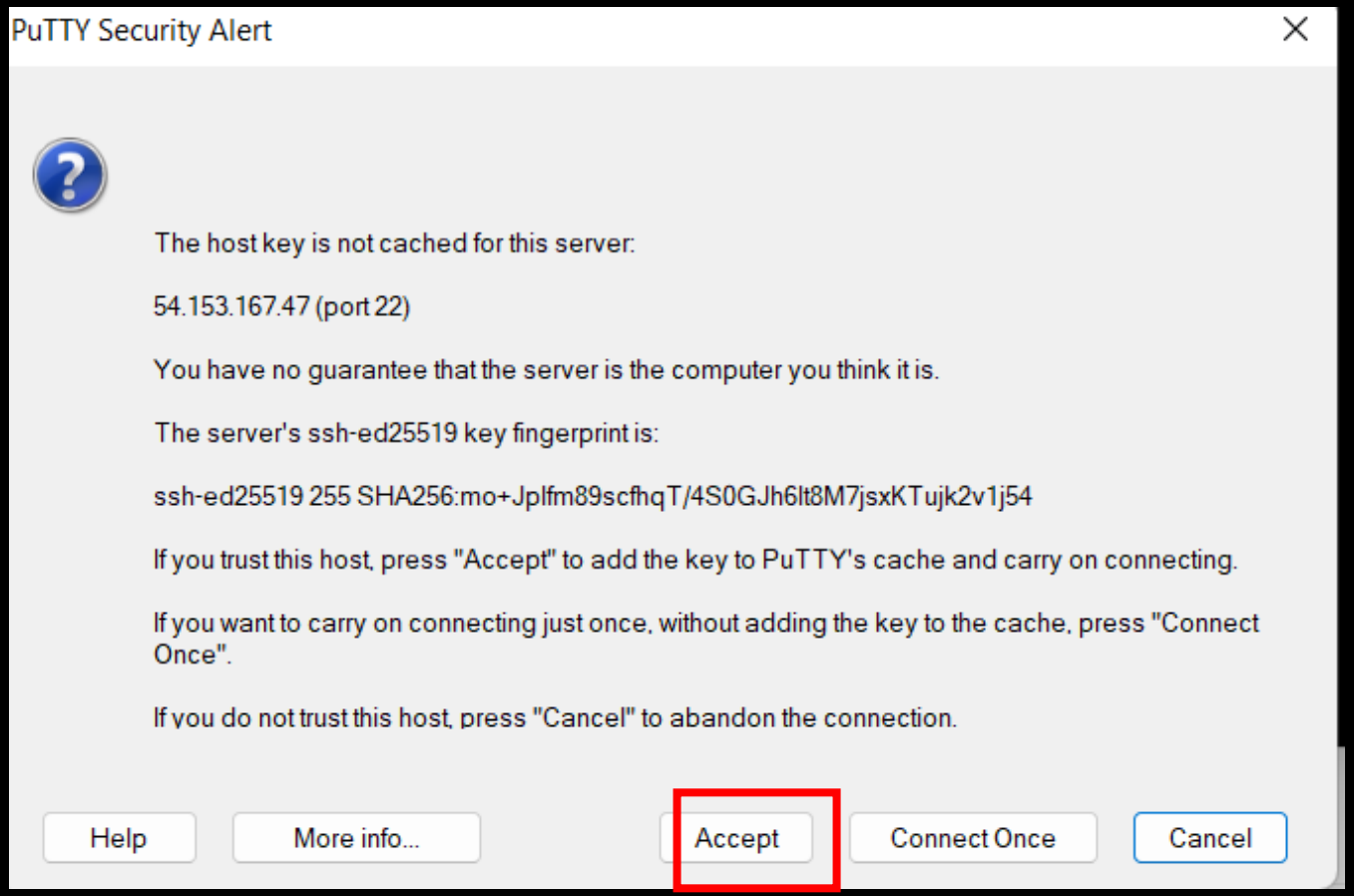
4. Under Private key file for authentication, browse and select your .ppk file. 🔑




Select private key file
This PC › Downloads
Search Downloads
Organize
New folder
Pictures
cloud and interv
itvedant
Music
Screenshots
OneDrive
This PC
Desktop
Documents
Downloads
Name
Date modified
Type
Size
Today (1)
my-linux-key-for-putty.ppk
06-07-2026 15:00
PPK File
Last week (5)
7.ppk
02-07-2026 17:49
PPK File
s3_static_website_hosting
03-07-2026 10:26
File folder
WEATHER_APP
02-07-2026 12:47
File folder
password-genrator
02-07-2026 12:47
File folder
product
30-06-2026 16:04
File folder
File name:
PuTTY Private Key Files (*.ppk)
Open
Cancel

5.Click to open.




PuTTY Security Alert
The host key is not cached for this server:
54.153.167.47 (port 22)
You have no guarantee that the server is the computer you think it is.
The server's ssh-ed25519 key fingerprint is:
ssh-ed25519 255 SHA256:mo+Jplfm89scfhqT/4S0GJh6lt8M7jsxKTujk2v1j54
If you trust this host, press "Accept" to add the key to PuTTY's cache and carry on connecting.
If you want to carry on connecting just once, without adding the key to the cache, press "Connect Once".
If you do not trust this host, press "Cancel" to abandon the connection.
Help
More info...
Accept
Connect Once
Cancel

6. Hit Open ,You’re almost there ✸



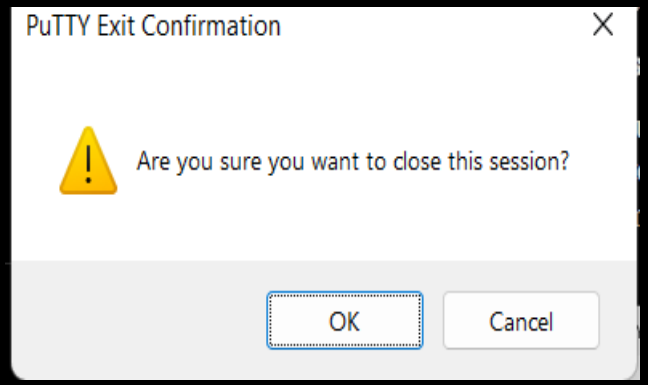
7. You’re officially inside your shiny new Linux machine! 💻✨



- CLEANUP AND EXIT.

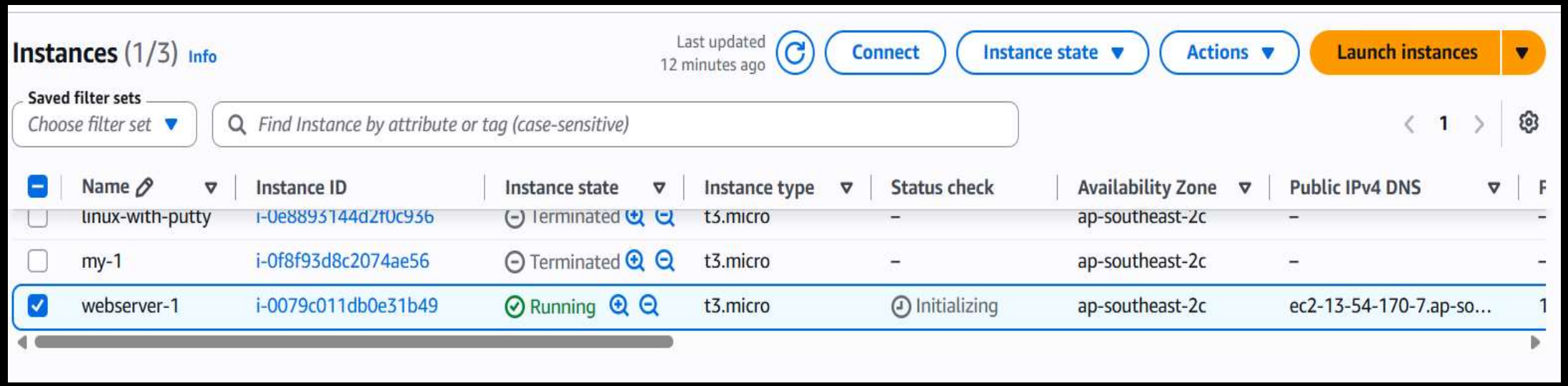
1. **Exit the session cleanly:**

- If you're done, you can close the PuTTY window or select the "X" mark in the top-right corner to terminate the session. Confirm the action if prompted.

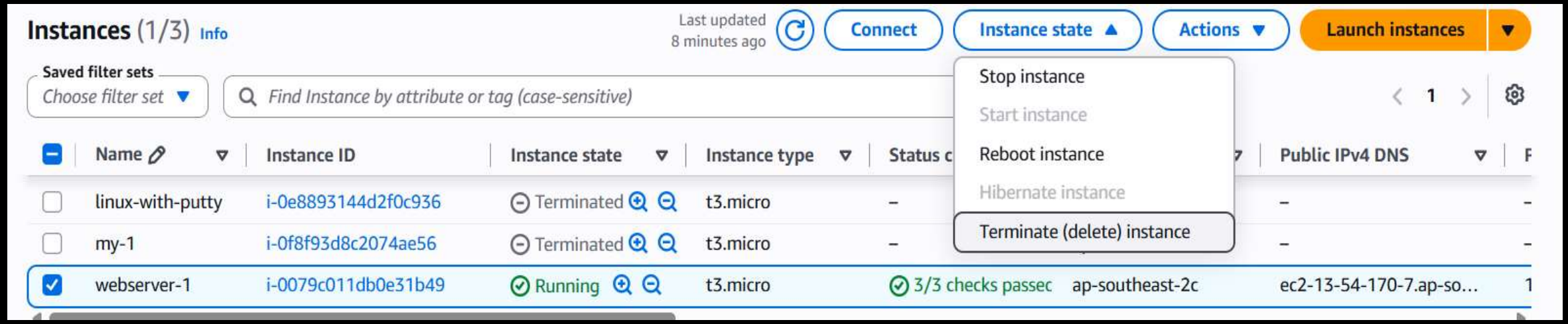


2.Delete the EC2 Instance:

- Go back to your **AWS Management Console**.
- Navigate to **EC2** and find the instance you created earlier.



- Select your **EC2 instance** and click **Terminate** from the click the Instance statae select the terminate option.



- Confirm that you want to terminate the instance. It will stop running, and you won't be charged for the instance anymore.

